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Dynamic Calculation Method of Interwell Connectivity Based on the Attention Mechanism

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ABSTRACT

The analysis of interwell connectivity plays a pivotal role in the optimization of the injection-production relationship, the formulation of an oilfield development plan and the description of the remaining oil distribution. Nevertheless, conventional techniques for determining interwell connectivity, such as tracer analysis, interference well testing, and numerical simulation, are characterized by high costs and low efficiency. To address this challenge, an increasing number of artificial intelligence methods have been proposed. However, these methods often necessitate the availability of a substantial quantity of labeled data to facilitate model training, and sensitivity analysis is required to ascertain the interwell connectivity of the model training results. Furthermore, the majority of existing methods are unable to perform dynamic analysis of interwell connectivity. This paper proposes a dynamic calculation method of interwell connectivity based on an attention mechanism to address the aforementioned issues. The interwell connectivity is taken as the intermediate parameter of the neural network, with the interwell connectivity between each injection well and the current production well in the well group inverted by predicting the daily fluid production of each production well in the well group. The experimental results demonstrate that the interwell connectivity calculations presented in this paper are in accordance with the actual operational conditions observed in the oil field and the reservoir expertize of industry professionals. This indicates that our approch can effectively support dynamic decision-making in the production process of oil reservoirs.

1 | Introduction

In water-flooded reservoirs, oil production can be driven by injecting water into some of these wells. Interwell connectivity describes the water flow path and transport capacity between these injection and production wells [1]. Figure 1 depicts the creation of a direct fluid connection between wells by injecting water into one well to increase reservoir pressure and allow surrounding oil to flow to other wells for collection. The importance of interwell connectivity for optimizing injection and production measures and improving the recovery of water-flooded reservoirs cannot be overstated [2–4]. In addition to its impact on the rationality and efficiency of the later development of the oilfield, interwell connectivity directly affects the effective exploitation of the remaining oil [5]. Consequently, analyzing and understanding interwell connectivity is essential for the development and management of water-flooded reservoirs.

In recent years, research into the connectivity of reservoir wells has focused on both static and dynamic aspects [6]. On the one hand, static connectivity studies are typically conducted to assess the initial state of a reservoir before development. The study of static connectivity is important for determining reservoir size, optimizing well design, predicting production capacity, and designing

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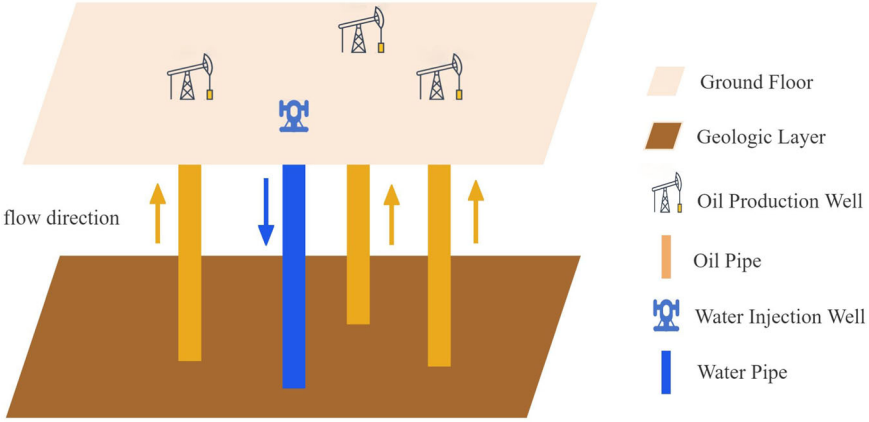

FIGURE 1 | Diagram of interwell connectivity.

fracturing programmes. Currently, several methods have been developed both domestically and internationally to describe and evaluate the static connectivity of reservoirs. These methods include exploration logging, seismic analysis, reservoir parameter comparison, and geological modeling [7]. Although the majority of techniques employed to assess interwell connectivity via static methods are relatively straightforward, they are unable to accurately portray the fluctuations in the migration of oil and gas within the reservoir. On the other hand, the study of dynamic connectivity provides a more accurate understanding of the flow path and velocity of oil and gas at different locations and times within the reservoir, as well as a timely understanding of the evolutionary law, even as the reservoir changes. Dynamic studies can more accurately predict the reservoir's production potential, optimize production plans, reduce production costs, and improve field recovery and production. Currently, several methods have been developed both nationally and internationally to describe and evaluate the dynamic connectivity between wells. These methods include geochemical analysis, tracer analysis, interference well testing and numerical simulation [8, 9]. However, despite their usefulness, these methods still have limitations. The tracer monitoring method accurately reflects subsurface well connectivity, but it is costly and requires long-term monitoring at the wellhead to avoid omissions [10, 11]. It also has environmental requirements that make it unsuitable for large-scale application. Furthermore, interference well testing directly reflects interwell connectivity, but is also costly and can disrupt the normal production schedule of the field. Reservoir numerical simulation can dynamically track reservoir flow [12], but it requires high-quality geological data and long-term manual adjustment [13]. It is important to note that this method should only be used when the necessary data is available and the manual adjustments are feasible.

The development of artificial intelligence technology has led to significant advancements in various fields, with artificial intelligence methods demonstrating remarkable capabilities in data utilization, complex relationship identification, and strong applicability and versatility. Consequently, artificial intelligence methods have emerged as a superior approach for addressing challenges encountered by traditional methods. In light of this, an increasing number of artificial intelligence methods have been proposed in recent years for the analysis and calculation of reservoir interwell connectivity. These methods can be broadly categorized into two groups: direct prediction of interwell connectivity and indirect inversion of interwell connectivity through prediction.

The initial approach is to utilize the interwell connectivity values calculated by the traditional method as the prediction target, thereby enabling the direct prediction of interwell connectivity. Cao et al. proposed an automatic identification method based on decision trees, which employs the CART algorithm to construct a decision tree, analyses oil field development data to extract potential knowledge, and establishes a direct mapping relationship between logging data and interwell connectivity of injection and production wells [14]. As historical data is often unavailable, the use of interwell connectivity as a training model label may result in a model with limited generalization capabilities.

The second method is to invert the interwell connectivity and indirectly predict the interwell connectivity coefficient by predicting the related indicators that reflect the interwell connectivity, such as permeability, porosity and dynamic production parameters. The research methodology is as follows: Panda and Chopra trained artificial neural network (ANN) was used to estimate well interactions between different well pairs [15]. The estimated well interactions were then used to determine the presence of heterogeneity, such as faults, cusp-off points, and regional permeability trends. Liu et al. proposed a method for estimating interwell connectivity based on bottomhole pressure data, employing ANN techniques [16]. This method uses bottom-hole pressure data as label data, although bottom-hole pressure data are often not directly available. Cheng et al. proposed a method for interwell connectivity analysis based on ANN [17]. In this method, the Long Short-Term Memory (LSTM) method is applied to the integrated reservoir model, and the mapping function between the producing well and the surrounding injection well is defined according to the historical water injection and liquid production data. Du et al. proposed a method to use a 3D convolutional neural network (CNN) to build a deep learning model and incorporate dynamic production data to invert interwell connectivity [18]. Cheng et al. utilized a substantial quantity of historical injection and production data from a synthetic reservoir model and employed an LSTM neural network to establish a mapping relationship between the producing well and the surrounding injection well. Subsequently, the interwell connectivity was evaluated on the

basis of the generated LSTM model utilizing the GSA method based on the Extended Fourier Amplitude Sensitivity Test (EFAST) [19]. A method combining Kalman filter and ANN was established on the basis of production dynamic data of injection and production wells. According to the sensitivity analysis rule, the sensitivity coefficient of the production well to the injection well was obtained, which was used to characterize the connectivity of the production well and the injection well. Although the aforementioned methods identify relatively well-obtained data as the labels for the training model, numerous studies still necessitate the combination of formulas or the performance of sensitivity analysis to calculate the interwell connectivity coefficient. This renders the calculation of the interwell connectivity coefficient both time-consuming and inaccurate.

In recent years, with the development of graph neural networks, some scholars have proposed the use of graph neural networks to analyze the interwell connectivity. For example, Zeng et al. utilized the adjacency matrix of a graph to represent the well pattern and proposed a Transformer-based collaborative task neural network from the perspective of water injection well groups and production well groups to learn for interwell connectivity evaluation [20]. Although this model is capable of calculating the dynamic connectivity coefficient, it is only able to evaluate the interwell connectivity of each month, and is therefore unable to evaluate the interwell connectivity of each day. Huang et al. proposed an improved graph neural network to simulate the interaction process of real flow laws between wells, with the aim of inverting the dynamic connectivity within each layer of injection and production wells [21]. Nevertheless, graph neural networks typically necessitate a substantial quantity of data for training, which consequently diminishes their computational efficiency.

In fact, the interwell connectivity is implied in the production dynamics of injection wells and production wells, which is often manifested as the degree of influence of injection wells on the daily fluid production of production wells. The greater the degree of this effect, the higher the interwell connectivity. Considering the extensive application of attention mechanism [22] in oil and gas field production [23–25], its working principle reveals that the attention weight in the attention mechanism can be used as an effective representation of interwell connectivity. Therefore, we propose a dynamic calculation method for interwell connectivity based on the attention mechanism. This method takes the well group as the unit and the daily fluid production of the current production well as the label, establishes the functional relationship between the multiple injection wells in the well group and the current production well itself and the label, respectively, and extracts the weight factor of the influence of the injection well on the label in the attention calculation to characterize the interwell connectivity. The model label is the daily fluid production of the production well, which is a good production data to obtain, and the model can accurately and quickly learn the interwell connectivity hidden in the production dynamics.

The remaining sections of this paper are as follows. First, in Section 2. *Methodology*, we outline the approach taken in this paper. This paper mainly includes the data and processing methods used in this paper, and puts forward the dynamic calculation model of interwell connectivity in this paper. The model mainly uses injection and production data to fit the daily fluid production by extracting relevant features, and obtains the intermediate results obtained by the model in the attention mechanism as the interwell connectivity results. Secondly, in Section 3. *Experiments*, We introduce the whole experimental process, including experimental details and experimental results, and verify the accuracy of our method from model fitting and reservoir expert experience. Finally, in Section 4. *Conclusion and Future Work*, we conclude the paper and outline future work.

2 | Methodology

This section consists of two Sections 2.1. and 2.2. In Section 2.1, we first introduce the data used in this paper, and also describe the problem, mainly introducing the composition, processing, and feature selection of data. In Section 2.2., we first describe the overall structure of the model, and then introduce the MFDUF and ACAM in detail.

2.1 | Problem Statement

In this paper, we use historical production data, well group relationship data, and geological description data of injection-production wells from a water-driven reservoir located in southern China. Specifically, the production data used are day-level data, including nine features of injection wells and 17 features of production wells. The well group relation data used include the number of injection wells and production wells in the well group, the corresponding well identification, well head coordinates, and so on. The geological description data used are monthly data, including vertical thickness, effective permeability, and porosity.

Taking into account the time delay of water injection in injection wells, this paper adopts the sliding window mechanism to obtain the production data of injection wells. At the same time, considering geological parameters can reflect interwell connectivity to a certain extent, so we propose to integrate geological parameters into the model to fine-tune the attention weight matrix. In the face of the above data, to solve the problem that the target calculation of interwell connectivity coefficient is difficult to obtain a large number of label data, and the dynamic interwell connectivity coefficient cannot be obtained, this paper designs a targeted algorithm to solve the problem. The following is the expression of the related function relationship of this method, as shown in Equation (1).

$$y = f(\boldsymbol{x_w}, x_{o_T}, x_g), \tag{1}$$

where the $\boldsymbol{x_w} = (x_{w_1}, x_{w_2}, x_{w_3}, ..., x_{w_T})$, T represents the time window size corresponding to the injection well, x_{w_i} represents the molding production data of the injection well on day i in the sliding window size of T, x_{o_T} represents the molding production data of the production well on day T, x_g represents the weight matrix calculated by the geological parameters of the month, y represents the daily fluid production of the well on day T, and f represents the model function established based on the attention mechanism.

TABLE 1 | Result of the feature selection.

Feature	Unit	Feature	Unit
Daily injection	m^3	Oil pressure	MPa
Wellhead injection pressure	MPa	Casing pressure	MPa
Water nozzle diameter	Mm	Production duration	H
Accumulate injection	10^4 m^3	Daily fluid production	m^3
Injection duration	H	Water ratio	/
Porosity	%	Oil nozzle diameter	mm
Permeability	D		

Before the data are trained, the data goes through a preprocessing step, including missing value input, outlier removal, feature selection, and normalization.

Data preprocessing is the first step to ensure the quality and reliability of data, and it is particularly critical to deal with missing values and outliers. In this study, the mean interpolation method based on historical data is used to fill missing values, and the distance-based KNN method is used to detect and deal with outliers. In addition, feature selection is also important for optimizing model performance, and in this study, a filtering method is used to select the most relevant feature subset, as shown in Table 1. Mutual information is used as the evaluation criterion to evaluate the importance of each feature, and a threshold is set for feature selection. Through the above preprocessing steps, it lays a solid foundation for subsequent data analysis and modeling.

Given the different units and ranges of different feature data, this paper adopts the max-min normalization strategy to map each data point to the interval [0,1]. Max-min normalization is a linear transformation method that compresses data into a certain range, usually [0,1] or [−1,1]. This approach not only unifies the dimension so that all features are present on a similar scale, thereby increasing the robustness and convergence rate of the model, but also preserves the distributional characteristics of the data and maintains the relative relationship between the data [26]. The mathematical expression of the normalization method is shown in Equation (2).

$$N_i = \frac{x_i - x_{\min}}{x_{\max} - x_{\min}}, \tag{2}$$

where N_i is the normalized value of the i-th data point, x_i is the original value of the ith data point, $x_{\min}$ is the minimum value in the original data set, $x_{\max}$ is the maximum value in the original data set.

2.2 | Model Design

2.2.1 | Overall Architecture

The basic structure of the dynamic calculation model of interwell connectivity based on the attention mechanism in this study is shown in Figure 2. As shown in Figure 2, the proposed model is mainly composed of two modules: MFDUF module and ACCAM module.

For injection wells and production wells in the same well group relationship, we assume that there is a connectivity relationship between them. Based on the above basic premise, the model we designed is well group as the unit, and the model can dynamically calculate the interwell connectivity coefficient between each injection well and the current production well.

Suppose a well group G contains n injection wells and m production wells. By inputting the dynamic production data of n injection wells with time window size T and the dynamic production data of current production wells on day T, the algorithm finally fits the daily fluid production on day T of current production through the MFDUF module and ACCAM module in turn. Through the attention mechanism, we can obtain the attention weight matrix of n injection wells and the current production well itself to the daily fluid production of the production well. The connectivity between the injection well and the production well is characterized by the attention weight value of the injection well to the production well, so as to obtain the interwell connectivity coefficient in this paper.

2.2.2 | MFDUF

Since the interwell connectivity is reflected in the connection between injection wells and production wells, it is necessary to fuse the feature information of both wells first. In this paper, the MFDUF module is designed to perform multi-source feature dimension unification and feature fusion. The input of MFDUF module includes the feature tensor of n injection wells in the same well group and the feature tensor of the current production well, which is composed of a double-branch structure. One branch is to fuse the two kinds of feature information in a dot product way, so that the model can learn the association and importance between different wells, so as to better capture the relationship between production wells and injection wells. The other branch fuses the two feature information in a concatenation manner, and the concatenated data retains all the information without losing any details.

This study utilizes a four-dimensional tensor to represent the time series data of the input model. The input data tensor can be expressed as $N \times W \times T \times F$, where N represents the number of samples, W represents the number of wells, T represents the time dimensions, and F represents the feature dimensions.

Specifically, the linear layer is first used to unify the input injection well feature tensor with the production well in the feature dimension. At the same time, the repeat vector layer is used to replicate the input feature tensor of the production well, so that it is consistent with the injection well in the time dimension. Then, Transpose operation is performed on the feature tensors of injection well passing through the linear layer, and Dot operation is used to fuse according to the feature dimension. The tensors after feature fusion represented the correlation between the feature tensors of injection well on day T and the feature tensors of production well on day T, respectively. At the same time, Concat operation is used to concatenate the features obtained in the

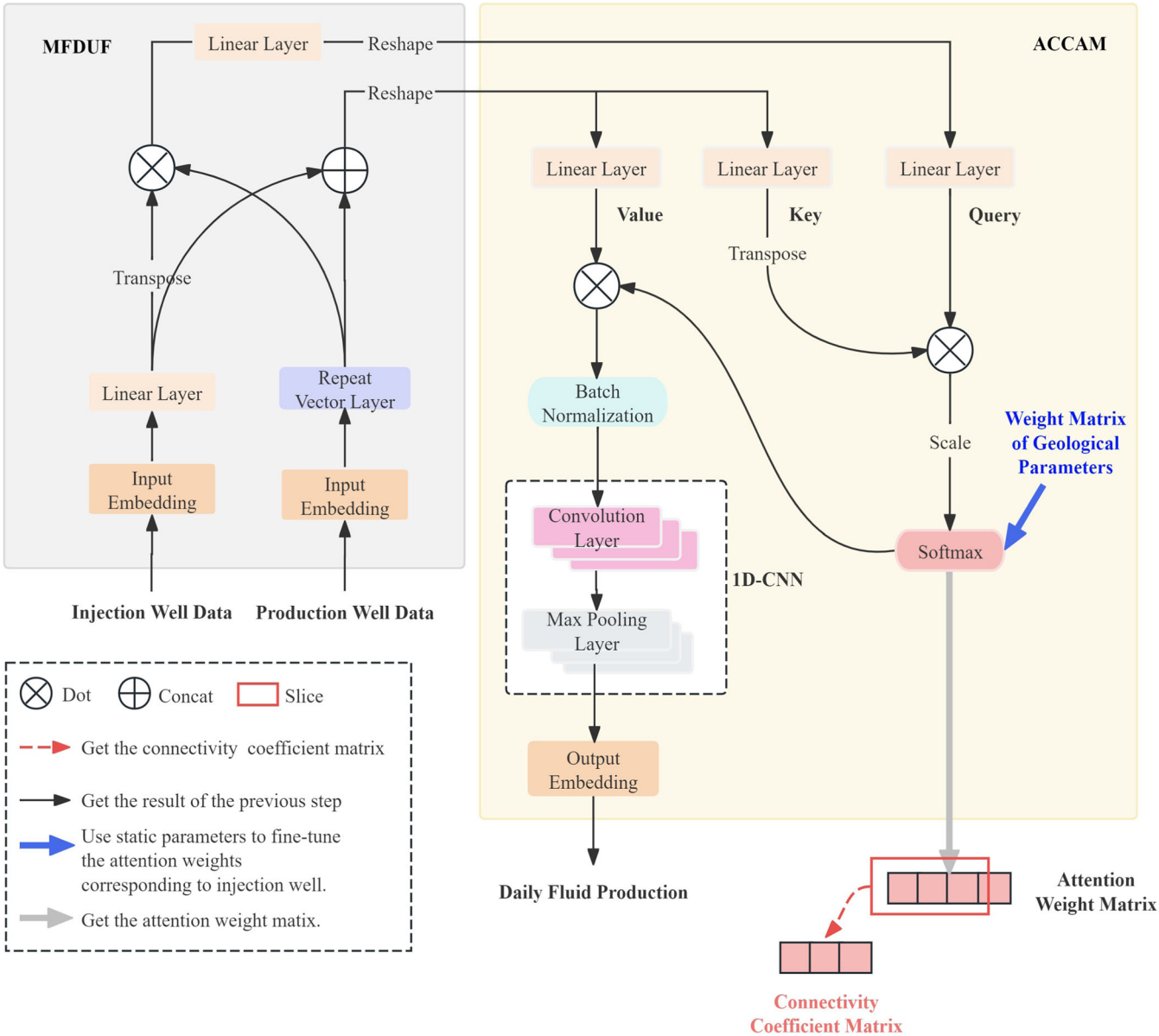

FIGURE 2 | Model structure diagram. The model is mainly composed of MFDUF module and ACCAM module, where MFDUF stands for multi-source feature fusion and extraction dimensional unity and fusion. ACCAM stands for adaptive computation of connectivity matrix based on the attention mechanism.

previous step in the quantitative dimension. Finally, to ensure the consistency of dimensions, the dot product fused tensors are processed by linear layer and Reshape operation, and the concatenated fused tensors are processed by Reshape operation.

2.2.3 | ACCAM

The ACCAM module has two functions: one is to realize the adaptive calculation of the connectivity coefficient based on the attention mechanism, and the other is to use the 1D-CNN to predict the daily fluid production.

Attention Mechanism Attention mechanism [22] is a technique widely used in the field of deep learning [27], which is inspired by human cognition of different degrees of attention to information. This mechanism improves the model's ability to represent the input by allowing the model to focus on key parts when processing sequence data. In traditional neural networks, each input element contributes equally to the output, while the introduction of attention mechanism changes this fixed weight setting. The core idea is that at each step of processing the input sequence, the model dynamically adjusts the degree to which it focuses on different parts, thus paying more attention to the information relevant to the current task.

As schematically indicated in Figure 3, the attention mechanism is composed of three main components: Query, Key, and Value. Query vectors represent locations or parts of the model's attention, which can be hidden states or other representations of the previous layer in the model. Key vectors represent locations or parts of the input sequence that the model wishes to focus on. The weight of attention is determined by calculating the dot product between the query vector and the key vector,

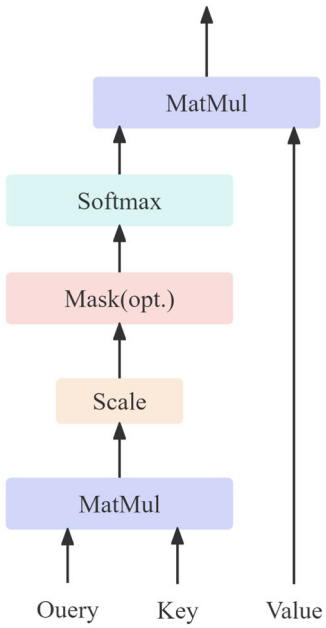

FIGURE 3 | Schematic of the attention mechanism.

which measures their similarity. The value vector contains information extracted by the model from the input sequence. The model uses a weighted sum of the value vectors as input, with the weights determined by the attention mechanism. information that is pertinent to the current task.

Equation (3) can be illustrated the attention mechanism very well. attention mechanism computes the weighted sum of the value vectors based on the compatibility between the query and the keys.

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V, \tag{3}$$

where Q is the query matrix of size $N \times d_k$, representing the information of current position or task, where N is the number of queries and d_k is the dimensionality of the query vectors. K is the key matrix of size $M \times d_k$, representing the information of all positions or tasks, where M is the number of keys and d_k is the dimensionality of the key vectors. V is the value matrix of size $M \times d_v$, representing the value information of all positions or tasks, where d_v is the dimensionality of the value vectors. Softmax function normalizes the scores obtained from the dot product of each query and key, ensuring that the sum of attention weights for each query over all keys is equal to 1. $\sqrt{d_k}$ is a scaling factor applied to the attention scores to prevent gradients from vanishing or exploding.

The calculation of attention weights usually involves calculating the similarity between the query vector and the key vector, converting the similarity into weights, and applying the weights to the value vector. Different similarity calculation methods and weight transformation methods lead to different types of attention mechanisms. Common attention mechanisms include scaled dot-product attention, multi-head attention, and self-attention. The dot-product attention is used in this paper. It calculates the dot-product of query vector and key vector, and adjusts with scale factor to get the similarity. Finally, Softmax function is used to get the attention weight. In sequential tasks, the attention mechanism enables the model to dynamically focus on different parts of the input sequence at different time steps or locations through the combination of queries, keys, and values. By calculating similarity and applying the Softmax function, the model can learn to assign different attention weights to different parts of the input. This flexibility allows the model to better capture long-term dependencies and important information in the sequence. In conclusion, the attention mechanism improves the ability of deep learning models to process sequence data by introducing dynamic attention mechanisms into different tasks, and plays an important role in improving the performance of various tasks.

CNN As an outstanding representative of deep learning, CNN [28] has achieved remarkable success in the field of computer vision [29–31], and it also shows excellent potential in the field of time series analysis [32, 33]. When dealing with time series data, CNN brings new ideas and effects for sequence modeling and prediction tasks by virtue of its unique structure and feature extraction ability [34].

The structure of CNN mainly includes convolutional layer, pooling layer, activation function and fully connected layer. The convolutional layer captures the local features in the sequence through convolution operation, and the pooling layer helps to improve the generalization of the model by reducing the data dimension and retaining key information. The introduction of downsampling and activation functions, as well as the fully connected layer to map features to the final output, together form the basic architecture of CNN [35].

Convolution is the core operation of CNN. By learning the convolution kernel, the network is able to recognize various features of the image, such as edges and textures. Multiple convolution kernels capture features of different scales and complexity at the same level, providing richer expression ability for the model. The pooling layer reduces the spatial dimension of the feature map and the computational complexity through operations such as Max pooling and average pooling, while retaining key information. In addition, the CNN uses a hierarchical structure to gradually extract the abstract features of the image through multi-layer convolution and pooling. The low level captures details such as edges and textures, and the high level learns more complex semantic information. Through the design of parameter sharing, CNN uses the same convolution kernel to extract features at different positions, reduces the number of parameters, and improves the generalization of the model.

CNN also has good transfer learning ability in time series prediction. Fine-tuning on the basis of the pre-trained model enables it to adapt to new time series tasks, and performs well especially in the case of limited data. The operation of 1D Convolution is shown in Equation (4).

$$y[n] = (x * w)[n] = \sum_{m=0}^{M-1} x[m] \cdot w[n-m], \tag{4}$$

where $x[n]$ is the input sequence, $w[n]$ is the weights of the convolutional kernel (filter), and M is the length of the convolutional kernel.

Detailed Introduction to ACCM We introduce the attention mechanism into the ACCM module for the following reasons: First, the attention mechanism allows the model to dynamically adjust the weight of information transfer between different wells according to actual needs, without the restriction of fixed structure, thus improving the model's ability to model complex dependencies and making it more adaptable to changing reservoir conditions. Secondly, the attention mechanism can adaptively focus on the relationship between different wells without relying on a fixed adjacency matrix, so it is more flexible to adapt to the actual well connectivity situation and reduces the dependence on specific assumptions. In addition, the attention mechanism is able to assign different weights to the relationships between different wells, so as to more accurately capture the complex relationships between wells and avoid possible errors caused by simplified processing. Finally, the weight distribution of the attention mechanism is interpretable, which can clearly show the attention degree of the model when modeling the connectivity between different wells, which helps us understand the mechanism behind the model decision and improves the interpretability of the model.

The attention mechanism used in the ACCAM module improves the performance of the entire neural network in two ways. Firstly, it helps to reduce the computational complexity. By focusing attention on important parts of the input data, neural networks can converge faster and reduce computational costs. Secondly, the attention mechanism helps to improve the generalization ability of the model. By capturing the association between different input data, the model can better adapt to various scenarios, thus improving its performance in practical applications.

In the ACCAM module, the feature tensor obtained from the first branch is used as the *Value* and *Key* in the attention mechanism after passing through the linear layer. This process can be expressed as follows:

- Value = linear (Feature Tensor from MFDUF module1)
- Key = linear (Feature Tensor from MFDUF module1)

Here, the role of the linear layer is to make dimensional transformations and parameter adjustments to the feature tensor to better capture important information about the input data in the attention mechanism. Next, we set the Query in the attention mechanism to the feature tensor obtained through the MFDUF module of the second branch. This process can be expressed as:

- Query = linear (Feature Tensor from MFDUF module2)

Again, the linear layer plays the role of adjusting the dimensions and parameters here so that the Query can effectively capture other important information of the input data. The core idea of attention mechanism is to obtain a new feature tensor by weighted summation of Value according to the similarity between Query and Key. This new feature tensor can be expressed as follows:

- Attention-weighted Feature Tensor = softmax (Query $*$ Key^T) $*$ Value

Here, the softmax function converts the similarity between the Value and the Key into a probability distribution to assign weights to the different elements of Value. These weights are subsequently used in a weighted summation to get new feature tensors that are more representative.

Considering the time delay of water flow from injection well to production well, it is affected by multiple factors of underground reservoir, such as porosity, permeability and formation thickness, and these geological parameters will change with time. Therefore, learning these underlying patterns from dynamic production data alone is challenging. In view of this, this paper not only uses the sliding window mechanism to obtain the production data of the injection well to retain the important information reflected in the time series of the injection well, and sets the size of the sliding window as a hyperparameter T. In addition, considering the influence of geological parameters (porosity and permeability) on connectivity, this paper proposes that after calculating the attention weight in the Softmax layer of this module, The geological parameter weights are used as the hyperparameters of the model, and the attention weight matrix corresponding to the injection well is fine-tuned. By comparing the weight values of different injection wells, we can intuitively understand which injection wells are more connected to the current production well, so as to optimize the water injection strategy and improve the recovery.

When performing feature extraction, 1D-CNN pays attention to the local correlation in the data. For data such as daily fluid production, there are usually some local patterns and trends. 1D-CNN can effectively capture these local features, thereby improving the accuracy of prediction.

In this paper, we use the 1D CNN, it is a variant of CNNs specifically designed for sequence data, such as time series or text data [36]. As shown in Figure 4, the architecture of 1D CNN is similar to that of traditional CNNs, but they operate on 1-D input data. The convolutional layer is the core component of 1D CNN, and sigmoid is used as the activation function in this paper. Equation (5) demonstrates that the Sigmoid function can map any input to a value between 0 and 1, making it useful for tasks requiring normalized output. Following the convolution layer, the data is subjected to max pooling, a downsampling technique that reduces dimensions while preserving key features. Max pooling reduces computational complexity while preserving important information, preventing overfitting and speeding up training.

$$f(x) = \frac{1}{1 + e^{-x}}, \tag{5}$$

where x is the input to the function, e is the base of the natural logarithm (Euler's number), $f(x)$ represents the output of the function, which is a value between 0 and 1.

As shown in Figure 2, finally, the feature tensor calculated by the attention mechanism will be fitted to the daily fluid production through the batch normalization layer and the 1D-CNN network (composed of multi-layer one-dimensional convolution and Max pooling layer) to realize the prediction of the daily fluid production by the model.

The model is trained by backpropagation and the connectivity matrix parameters are trained iteratively to make the model fit the daily fluid production more accurately, and at the same

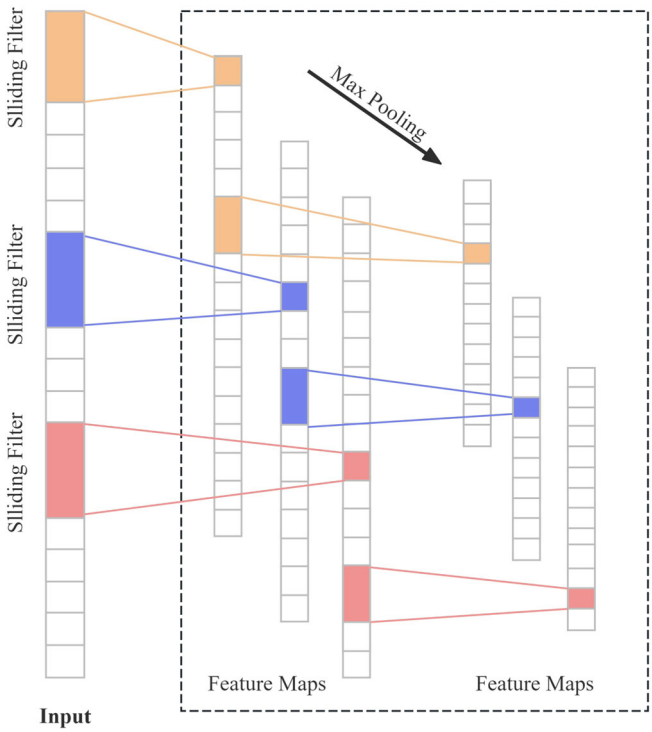

FIGURE 4 | Schematic of the 1D-CNN.

time, make its characterization of the connectivity relationship between water injection wells and oil recovery wells (i.e., the water injection allocation benefit) more accurate.

3 | Experiments

This section consists of Sections 3.1 and 3.2. In Section 3.1, we deeply expounds the key parameter settings in the experiment process and the performance measurement indicators of the model. In Section 3.2, we verified the effect of the model by comparing the fitting effect of the observation model with the empirical understanding of reservoir experts.

3.1 | Experimental Details

3.1.1 | Training Setting

Taking well group G1 as an example, the well group contains one injection well I1, three production wells P1, P2, and P3. The production wells in the well group are modeled successively: First, the dynamic production data and geological data of the injection well and the current production well in the well group for nearly 5 years are preprocessed. Secondly, this paper makes an input module data set based on the sliding window mechanism for the pre-processed historical production data, where the size of the sliding window $T = 90$. This is due to the ability of the sliding window mechanism to convert the time series data into a fixed length sample sequence, which helps capture the time dependence, extract the sequence features, and enable the model to better understand and process the time series data. Finally, the data set obtained by the sliding window mechanism is divided into the training set (2100 samples) and the test set (900 samples) according to the ratio of 7:3. As shown in Table 2, the data formats for model input and output are shown. The early stop strategy is adopted in the training process of the model, which can prevent the model from overfitting or early convergence during the training process, so as to improve the generalization ability of the model. In addition, the optimization algorithm adopted is Adam. By combining the characteristics of momentum and adaptive learning rate, Adam algorithm can improve the convergence speed, improve the training stability, and reduce the number of hyperparameters that need to be manually adjusted when training deep learning models. The experimental parameter Settings in the training process are shown in Table 3.

3.1.2 | Measure Metric

Model evaluation is a crucial step in machine learning. It is important not only to verify the accuracy of the model but also to optimize its performance. By analyzing the results of the model, we can understand its performance and adjust the model parameters and structure accordingly. Among the many evaluation metrics, mean squared error (MSE) has gained attention due to its wide range of applications. Equation (6) shows that MSE evaluates the accuracy of the model's predictions by calculating the squared difference between the predicted and true values. A smaller MSE value indicates better model prediction performance. The back-propagation algorithm is typically used to optimize model parameters during training. After each training session, the model's MSE value is calculated to evaluate its effectiveness. Through iterative training and gradual reduction of the MSE value, the model's performance can be improved.

$$\mathrm{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2, \tag{6}$$

where n is the number of samples, y_i is the actual values, $\hat{y}_i$ is the predicted values.

The coefficient of determination (R^2) is also used to evaluate the fit of the regression model. The R^2 indicates how well the model fits the data, that is, how well the model explains the variance of the observed data. In regression analysis, the coefficient of determination is an important statistic that measures the extent to which the independent variable explains the variation in the dependent variable. Specifically, the coefficient of determination can be defined as the proportion of the variance of the observed data that is explained by the model, which is usually expressed as a numerical value between 0 and 1. In general, the closer the R^2 value is to 1, the better fit the model is to the data, so in this case the model is better able to explain the variation in the dependent variable. In other words, a larger R^2 value means that the model is able to explain more of the observed variance, and the difference between the predicted results and the actual observed values is smaller. The coefficient of determination is calculated as shown in Equation (7).

$$R^2 = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2}, \tag{7}$$

where y_i is the observed actual value of the dependent variable, $\hat{y}_i$ is the predicted value of the dependent variable by the regression model, $\bar{y}$ is the mean of the dependent

TABLE 2 | Data format in the model.

	Parameters	Data format
Input	Injection well	$1 \times 90 \times 6$
	Production well	$1 \times 1 \times 5$
Intermediate input	Weight matrix of the geological parameters	1×1
Intermediate output	Connectivity coefficient matrix	1×1
Output	Daily fluid production	1×1

TABLE 3 | Parameters for model training.

Parameters	Value
Learning rate	0.001
Batch size	64
Epoch	200
T	90
Number of samples	3000

variable, and the summation symbol $\sum$ means summing over all samples.

The references for the above indicators are [37–39].

3.2 | Experimental Results and Verification

3.2.1 | Model Fit

This paper uses a rigorous evaluation method to verify the effectiveness of the prediction model by testing the degree of fit between the prediction model and the actual data. As shown in Figure 5, the historical fitting curve between the predicted value and the real value of daily fluid production has been shown in detail. Meanwhile, the coefficient of determination (R^2) has also been introduced as an indicator lto evaluate the fitting effect of the regression model. After careful analysis, we found that in this well group, the R^2 value of the model corresponding to each production well exceeds 0.97, which indicates that the model fits well. Therefore, we have reason to believe that the interwell connectivity coefficient calculated by this model is reliable.

3.2.2 | Field Experience of Reservoir Specialists

The connectivity between injection wells and production wells usually shows an increasing trend. This is due to the fact that the pressure in the underground reservoir will gradually decrease over time and the mobility of the crude oil will weaken during the process of oilfield extraction. Therefore, to maintain or increase crude oil production, water injection is often used to increase the pressure in the underground reservoir to promote the movement of crude oil to the well. When water is injected into an injection well, the water will diffuse around through pores and fractures in the reservoir. If there is connectivity between the injection well and the production well, then the injected water will gradually move toward the production well and push the surrounding crude oil toward the production well. As a result, the connectivity between the injection well and the production well increases over time, increasing the effective recovery of oil from the underground reservoir to the production well. Therefore, as oilfield production continues, the connectivity between injection wells and production wells tends to increase, which is one of the important strategies to improve oil recovery efficiency and maintain oilfield production. Therefore, we can observe whether the changes of the interwell connectivity coefficient at different times conform to the above rules, and take this as one of the methods to verify the effect of the dynamic calculation of the interwell connectivity coefficient of the model.

We respectively calculated the interwell connectivity coefficient of each production well and injection well in well group G1 at different times, and drew the corresponding curve graph. The experimental results are shown in Table 4. As can be seen, the interwell connectivity changes relatively smoothly within half a year, but all show an upward trend. From Table 4, it is easy to see that the connectivity between injection well I1 and production well P3 is the best, the connectivity between injection well I1 and production well P1 is good, and the connectivity between injection well I1 and production well P2 is poor. Therefore, the connectivity between injection well I1 and each production well is as follows: P3 > P1 > P2.

After statistical analysis, we introduce the connectivity coefficient c as the basis for the rank division. In view of the complexity of geological environment, this paper assumes that there is a connected relationship between the injection and production wells in the same well group. Based on this assumption, the connectivity relationship is subdivided into three levels: weakly connected (c ranges from 0.0 to 0.4), moderately connected (c ranges from 0.4 to 0.7), and strongly connected (c ranges from 0.7 to 1.0). Taking well group G1 as an example, the connectivity between I1 and P1 and P2 belongs to a medium level, while the connectivity between I1 and P3 reaches a high level.

Figure 6 is the understanding of the strong and weak relationship of interwell connectivity provided by reservoir experts. Experts believe that the strength of interwell connectivity of G1 in June 2023 is as follows: the interwell connectivity between injection well I1 and production wells P3 and P1 is good, which is due to the rapid rise of water in production wells P3 and P1, while the interwell connectivity between injection well I1 and production well P2 is poor, because production well P2 basically does not see water. Therefore, there is uneven water flooding in well group G1 on the plane. Finally, experts gave the relationship between the effect strength of injection well I1 and each production well as: P3 P1 P2. Through comparison, it can be found that the connectivity

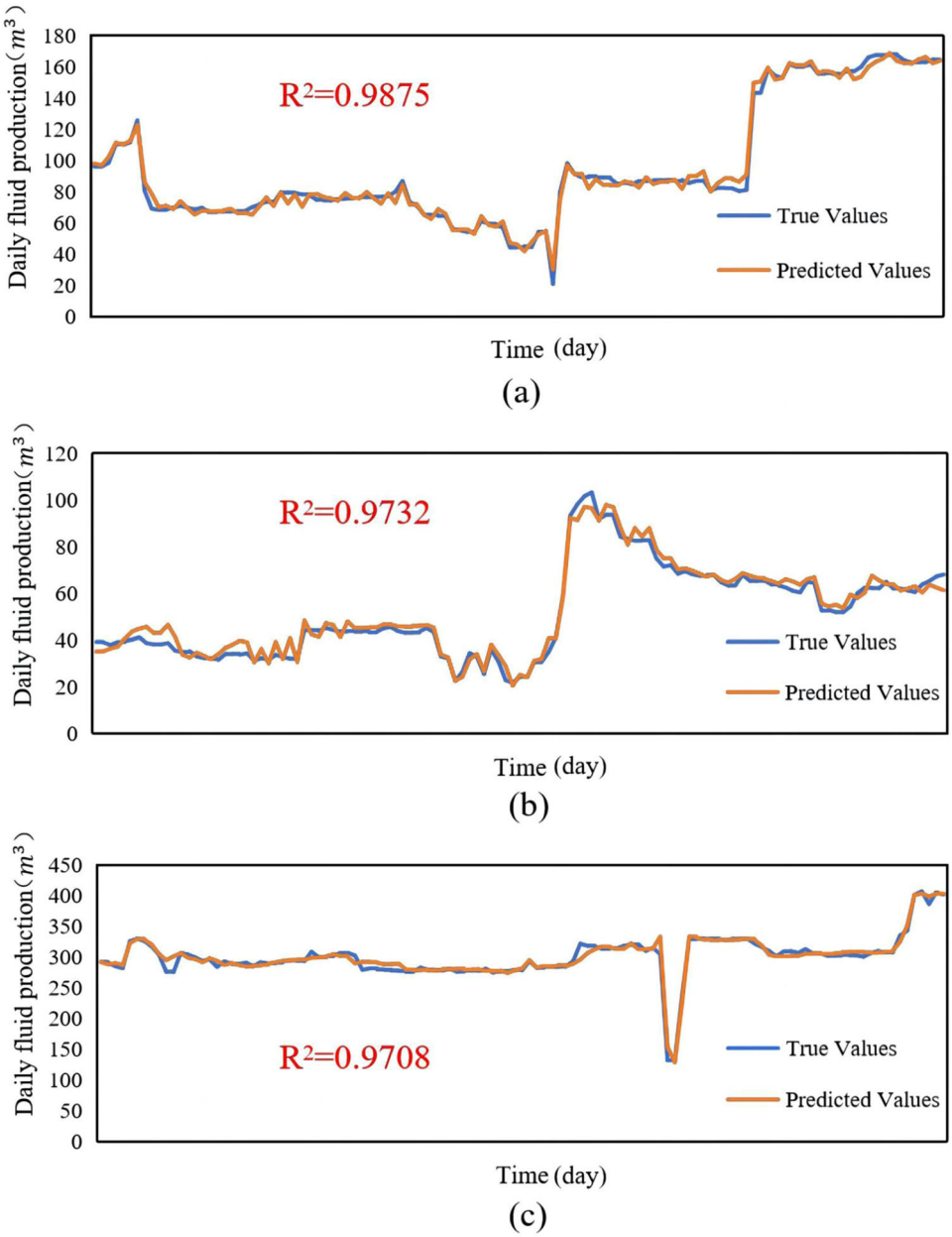

FIGURE 5 | Historical fitting results for individual production wells in well group G1, where figures (a)–(c) represent the historical fitting curves of production wells P1, P2, and P3, respectively.

coefficient results calculated by the model are consistent with the understanding of connectivity strength provided by experts, which further verifies the accuracy of the model.

4 | Conclusion and Future Work

In this paper, we successfully realize the dynamic calculation of interwell connectivity by introducing a method based on the attention mechanism. Compared with previous studies on interwell connectivity analysis, our method not only overcomes the problem that a large number of labeled data cannot be obtained, but also completes the acquisition of dynamically changing interwell connectivity coefficients. The intermediate results of the model obtained based on the attention mechanism are used as interwell connectivity coefficients to fill the gap in this field. In addition, compared with the use of graph neural network, the proposed method reduces the complexity of model calculation and greatly improves the efficiency of interwell connectivity analysis.

We propose a method for dynamic computation of interwell connectivity based on an attention mechanism. In this model, we propose two new modules: MFDUF and ACCAM. The connectivity between the injection well and the production well is characterized by extracting the attention weight of the daily fluid output of the injection well to the production well in the model. Furthermore, historical production data and geological data (porosity, permeability) are integrated to improve the model and make the prediction of interwell connectivity more accurate. The R^2 of the proposed model is more than 0.97 in terms of model fitting degree. To further verify the accuracy

TABLE 4 | Interwell connectivity coefficients corresponding to different dates in well group G.

Date	I1-P1	I1-P2	I1-P3
2023-01-08	0.521	0.444	0.550
2023-02-08	0.535	0.429	0.621
2023-03-08	0.649	0.440	0.704
2023-04-08	0.566	0.453	0.715
2023-05-08	0.566	0.453	0.700
2023-06-08	0.566	0.436	0.708

FIGURE 6 | Knowledge of reservoir experts on the strength of interwell connectivity of wells in well group G1. Thicker lines represent stronger connectivity, and vice versa.

of the model for the calculation of interwell connectivity coefficient, we compared the expert experience. Experimental results indicate that the interwell connectivity calculated by the model is consistent with the experience of reservoir experts.

By introducing the attention mechanism, we realize the adaptive calculation of interwell connectivity, which provides a powerful tool for the efficient development and production of oil and gas fields. The model's results have significant implications for oilfield management, particularly in optimizing water injection strategies and well placement. By leveraging the dynamic interwell connectivity information, oilfield operators can adjust injection strategies, enhance oil recovery, and improve overall field performance. Furthermore, combining interwell connectivity data with other production variables, such as production rates and water injection volumes, can guide more effective field development decisions. Looking to the future, with the continuous development of this technology and its deeper application in practice, we believe this method will play an increasingly important role in optimizing oilfield production strategies and improving operational efficiency in real-world oil and gas fields.

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